DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Course Details	
Class : III Year B. Tech Academic **Year:** 2025-26 **Course Code :23A02605** Branch: CSE,CSE-AI,AIML,DS,ECE,CIVIL/MECH	**Semester:** II **Course Title:** Renewable Energy Systems **Credits:** 3
Staff: Mr. R.Yedukondalu / Mrs.S.Suneetha	

QUESTION BANK
UNIT – I: SOLAR ENERGY

S.NO	QUESTION	CO	BL	MARKS
1.	Explain a) Extraterrestrial radiation b) Terrestrial Radiation c) Solar Constant d) Beam Radiation e) Diffuse Radiation	1	B2	10
2.	Define the earth sun angles 1. Latitude 2. Declination Angle 3. Hour angle 4. Zenith Angle 5. Inclination/ Altitude Angle	1	B1	10
3.	Explain the construction and working of Pyranometer with a neat sketch and also applications	1	B2	10
4.	Explain the construction and working Pyrheliometer with a neat sketch and also applications	1	B2	10
5.	a) Describe the Construction and operation of Flat plate Collectors.	1	B2	5
	b) what are the applications, advantages and disadvantages of Flat plate Collectors	1	B1	5
6.	(a) Describe the Construction and operation of Concentrating Collectors	1	B2	5
	(b) what are the applications, advantages and	1	B1	5

	disadvantages of Concentrating Collectors			
7.	Describe a)Local Solar time b) Sunrise c)Sunset and d) Day length	1	B2	10
8.	Explain about thermal energy storage system	1	B2	10
9.	Define the earth sun angles 1. Solar Azimuth Angle 2. Slope/Tilt angle 3. Surface Azimuth Angle 4. Incident Angle	1	B1	10
10.	Explain the construction and working of sunshine recorder with a neat sketch and also applications	1	B2	10

UNIT – 2: PV ENERGY SYSTEMS

S.NO	QUESTION	CO	BL	MARKS
1.	Explain the classification of Solar PV systems.	2	B2	10
2.	Explain single crystalline silicon PV cell along with advantages and disadvantages.	2	B2	10
3.	Explain multi crystalline silicon PV cell.	2	B2	10
4.	Explain amorphous silicon PV cell.	2	B2	10
5.	Define 1.PV cell 2. PV module 3.PV panel 4.PV array 5.Photo voltaic effect.	2	B2	10
6.	(a) Explain the electrical characteristics of PV Cell.	2	B2	5
	(b) Draw the Equivalent circuit of PV cell.	2	B2	5
7.	(a) Explain grid connected PV systems.	2	B2	5
	(b) Describe stand alone system configurations.	2	B2	5
8.	Explain distributed PV system.	2	B2	10
9.	Explain the V-I characteristics of series and parallel combination of PV cells	2	B2	10
10.	Describe the PV technologies other than silicon crystal cell and film cells.	2	B2	10

UNIT – 3: Wind Energy

S.NO	QUESTION	CO	BL	MARKS
1.	Explain the basic components of wind energy conversion system	3	B2	10

2.	Classify the wind energy conversion system	3	B2	10
3.	Explain the principle of wind energy conversion and wind power.	3	B2	10
4.	Explain in detail about Wind mill components of wind turbine	3	B2	10
5.	(a) Differentiate between the HAWT and VAWT	3	B2	5
	(b) Mention the considerations for Site selection of Wind mill	3	B2	5
6.	Analyze Aero dynamic forces acting on the blade in Wind mill.	3	B2	10
7.	With neat diagram explain the Darrieus type rotor .Mention its advantages, disadvantages.	3	B2	10
8.	Describe the types of vertical axis wind turbine.	3	B2	10
9.	With neat diagram explain the Savonius type rotor .Mention its advantages, disadvantages.	3	B2	10
10.	Describe the types of horizontal axis wind turbine.	3	B2	10

UNIT – 4: GEOTHERMAL ENERGY

S.NO	QUESTION	CO	BL	MARKS
1.	(a) What are the estimates of geothermal power in the earth	4	B1	5
	(b) What are the nature of geothermal fields of geothermal energy	4	B1	5
2.	With neat diagram, explain the process of heat extraction from hot dry rocks.	4	B1	10
3.	Describe various energy extraction technologies used with hydrothermal resources	4	B1	10
4.	(a) Discuss vapour dominated geothermal plant with a diagram	4	B2	5

	(b) What is the current status of geothermal energy in India.	4	B1	5
5.	Explain the following types of Geo-Thermal Resourses: a) Geo Pressured b) Magma	4	B2	10
6.	What are the advantages, disadvantages and applications of geothermal energy	4	B2	10
7.	List out various types of Geothermal resources and explain any of two geothermal resources	4	B2	10
8.	(a) What are the applications of geothermal energy	4	B1	5
	(b) What are the important aspects of geothermal energy	4	B1	5
9.	(a) What are the advantages and disadvantages of geothermal energy?	4	B1	5
	(b) Compare between Geothermal Power plant and Conventional thermal Power plant.	4	B2	5
10.	(a) Discuss vapour dominated geothermal plant with a diagram	4	B2	5
	(b) What is the current status of geothermal energy in India.	4	B1	5

UNIT – 5: GEOTHERMAL ENERGY

S.No.	QUESTION	CO	BL	MARKS
1	(a) Explain the working principle and operation of the tidal energy?	5	B2	5
	(b) What are the advantages and limitations of tidal energy	5	B1	5
2	(a) Explain the working of single basin tidal power plant	5	B2	5
	(b) Explain the working of Double basin tidal power plant	5	B2	5
3	(a) Explain the working principle and operation of the wave energy?	5	B2	5
	(b) What are the merits and demerits of wave energy	5	B1	5
4	Explain various types of wave energy conversion machines.	5	B2	10
5.	Discuss about the conversion technologies of bio-mass energy.	5	B1	10

6	Write the classification of bio-gas generation plants and mention its advantages and disadvantages.	5	B1	10
7	(a) Explain the choice of site selection to construct a bio-gas generation plant.	5	B2	5
	(b) Explain about the digester design consideration of biogas plant	5	B2	5
8	Explain the Polymer Electrolyte Membrane (PEM) Fuel Cell and its limitations?	5	B2	10
9	(a) Explain the construction and working of Hydrogen-Oxygen Fuel Cell (Hydrox Cell) or Alkaline Fuel Cell (AFC),	5	B2	5
	(b) What are the advantages, disadvantages and applications of Alkaline Fuel Cell	5	B1	5
10	(a) Explain the working Molten Carbonate Fuel Cell? Write the applications of fuel cells.	5	B2	5
	(b) What are the advantages, disadvantages and applications of Alkaline Fuel Cell	5	B1	5

SHORT QUESTIONS
(2 MARKS)
UNIT – I: SOLAR ENERGY

S.NO	QUESTION	CO	BL	MARKS
1.	What are the advantages and disadvantages of concentrating Collectors	1	B1	2
2.	Define Altitude angle, Zenith angle and Solar Azimuth Angle	1	B1	2
3.	Distinguish between flat plate and concentrating collectors	1	B2	2
4.	Distinguish between the beam and diffuse solar radiation	1	B2	2
5.	Define Latitude, hour angle, declination angel	1	B1	2
6.	Define Local solar Time	1	B1	2
7.	What are the advantages and disadvantages of Flat plate Collectors	1	B1	2
8.	What are the applications of concentrating collectors	1	B1	2
9.	Define the sensible heat and latent heat	1	B1	2
10.	What are the applications of pyreheliometer	1	B1	2

UNIT – 2: PV ENERGY SYSTEMS

S.NO	QUESTION	CO	BL	MARKS
1.	Define Photovoltaic cell.	2	B1	2
2.	Define Solar PV array.	2	B1	2
3.	What is Fill Factor.	2	B1	2
4.	Define Photovoltaic Effect.	2	B1	2
5.	Mention V-I Characteristics of parallel connected PV Cells.	2	B1	2
6.	What are the advantages and disadvantages of Mono-crystalline Silicon pv cell.	2	B1	2
7.	What are the Characteristics of Multi Crystalline Silicon pv Cell.	2	B1	2
8.	What are the Characteristics of Poly Crystalline Silicon PV Cell.	2	B1	2
9.	What are the advantages of organic solar cell.	2	B1	2
10.	Discuss different types of semiconductor materials used for solar cells.	2	B2	2
11.	Draw the equivalent circuit of PV cell.	2	B3	2
12.	What do you mean by Grid connected PV system.	2	B1	2

UNIT – 3 : WIND ENERGY

S.NO	QUESTION	CO	BL	MARKS
1.	Explain about cup type rotor.	3	B1	2
2.	Explain about pitch control system.	3	B1	2
3.	What is Yaw control.	3	B1	2
4.	What are the advantages of Vertical Axis Wind Turbine.	3	B1	2
5.	What are advantages of Savonius type rotor.	3	B1	2
6.	Differentiate between the HAWT & VAWT.	3	B1	2
7.	Write the Characteristics of savonius and Darrieus type rotor.	3	B1	2
8.	What are the disadvantages of Vertical Axis Wind Turbine.	3	B1	2

9.	What are the advantages and disadvantages of Wind Energy.	3	B1	2
10.	Explain about wind mill head.	3	B1	2

UNIT – IV: GEOTHERMAL ENERGY

S.NO	QUESTION	CO	BL	MARKS
1	What are the benefits of geothermal energy	4	B1	2
2	Why is geothermal energy a renewable resource	4	B1	
3	Where is geothermal energy available	4	B1	2
4	What are the environmental impacts of using geothermal energy?	4	B1	2
5	What are the different types of geothermal power plants	4	B1	2
6	What makes a site good for geothermal electric development	4	B1	2
7	What are the advantages of geothermal energy	4	B1	2
8	Do geothermal power plants produce emissions that lead to global warming	4	B1	2
9	What are the drawbacks and challenges of geothermal systems	4	B1	2
10	What is the role of magma in geothermal energy	4	B1	2
11	What factors affect the efficiency of geo thermal energy	4	B2	2

UNIT – V: MISCELLANEOUS ENERGY TECHNOLOGIES

S.NO	QUESTION	CO	BL	MARKS
1.	What is the principle of Tidal energy?	5	B1	2
2.	What are the type of tidal power plant?	5	B1	
3.	What are the advantage and disadvantages of tidal	5	B1	2
4.	What are the advantages and disadvantages of wave energy?	5	B1	2
5.	What are the types of wave conversion machines?	5	B1	2

6.	What are environmental effects of biomass/biofuels	5	B1	2
7.	What is the comparison of floating and fixed dome type biogas plants?	5	B1	2
8.	What is the fuel cell?	5	B1	2
9.	What are the types of fuel cells?	5	B1	2
10.	What are the limitations of alkaline fuel cell?	5	B1	2

Prepared by	**Verified by**	**Check and distribution of BL**	**Quality checked by**